

Vocabulary

Axis of Symmetry: The vertical line $x = h$ passing through the vertex of a parabola.

y-intercept: The point where a graph crosses the y-axis, found by substituting $x = 0$ into the equation.

We do together. Graph $y = (x - 3)^2 - 2$ using vertex, axis of symmetry, and y-intercept.

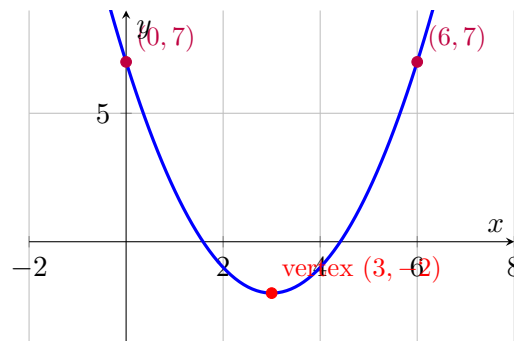
Step 1. Vertex: $(h, k) = (3, -2)$.

Step 2. Axis of symmetry: $x = 3$.

Step 3. y-intercept: substitute $x = 0$: $y = (0 - 3)^2 - 2 = 9 - 2 = 7$, giving point $(0, 7)$.

Step 4. Mirror point across $x = 3$: since $(0, 7)$ is 3 units left of the axis, the mirror point is 3 units right: $(6, 7)$.

Step 5. Plot $(3, -2)$, $(0, 7)$, and $(6, 7)$, then sketch the U-shaped curve.



$y = (x - 3)^2 - 2$ with vertex, y-intercept, and mirror point.

You Try. Solve each. Show every step, then name the answer.

- a) Find the vertex, axis of symmetry, and y-intercept for $y = (x + 2)^2 - 5$. Vertex $(-2, -5)$; axis of symmetry $x = -2$; y-intercept: $y = (0 + 2)^2 - 5 = -1$, point $(0, -1)$.

Summary (fill in): To graph a quadratic in vertex form, first plot the vertex (h, k) , then draw the axis of symmetry $x = h$. Find the y-intercept by substituting $x = 0$, then reflect that point across the axis of symmetry to get a matching point, and connect all the points with a smooth curve.